GLUEING AN INTEGRAL MORPHISM SO THAT OVER A POINT THE FIBRE BECOMES A SINGLE POINT WITH PRESCRIBED RESIDUE FIELD

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 7289-7346, immediately following the statement at 7273-7287.

Editorial note. Six cross-chapter pointers (Decent Spaces, More on Algebra, Morphisms, Morphisms of Spaces) are recorded in assembly.external_references.

Editorial note. The final finiteness statement is proved inside the span from the fibre product algebra being finitely generated over R.

Allowed standard prerequisites (editorial).

- Fibre products of algebras and their points. More on Algebra, situation more-algebra-situation-module-over-fibre-product and lemmas more-algebra-lemma-points-of-fibre-product and more-algebra-lemma-fibre-product-finite-type, cited throughout. Describes the fibre product algebra C' and its points, which is how the new fibre X'_y is controlled.
- Affine equivalence for algebras and modules. Morphisms, lemma morphisms-lemma-affine-equivalence-algebras with Morphisms of Spaces, lemma spaces-morphisms-lemma-pushforward, cited in the proof. Moves the construction between quasi-coherent sheaves and their global sections.
- Recall on decent spaces. Decent Spaces, remark decent-spaces-remark-recall, cited in the proof. Allows the residue field of the chosen point to be treated as a finite separable extension in the decent case.
- Integral morphisms are affine and closed. standard Stacks facts about integral morphisms used throughout the proof. Keeps the constructed factorization X -> X' -> Y integral over Y and lets the fibre condition be checked affine-locally.

Source reference. Original reference 03II: focused-seeds/geometry/S9-spaces-construct-glueing/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00IT: focused-seeds/geometry/S9-spaces-construct-glueing/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0B7J: focused-seeds/geometry/S9-spaces-construct-glueing/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof

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has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08KH: focused-seeds/geometry/S9-spaces-construct-glueing/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01SA: focused-seeds/geometry/S9-spaces-construct-glueing/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03M9: focused-seeds/geometry/S9-spaces-construct-glueing/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be an integral morphism of algebraic spaces over S. Let $y \in |Y|$ be a point which can be represented by a closed immersion $y : \mathrm{Spec}(k) \to Y$. Then there exists a factorization $X \to X' \to Y$ of f such that*

(1) *$X' \to Y$ is integral,*
(2) *$X \to X'$ is an isomorphism over $X' \setminus X'_y$,*
(3) *X'_y has a unique point x' with $\kappa(x') = k$.*

Moreover, if f is finite and Y is locally Noetherian, then $X' \to Y$ is finite.

Proof. By Morphisms of Spaces, Lemma 03M9 the sheaves $f_*\mathcal{O}_X$, $(X_y \to Y)_*\mathcal{O}_{X_y}$, and $y_*\mathcal{O}_{\mathrm{Spec}(k)}$ are quasi-coherent sheaves of $\mathcal{O}_Y$-algebras. Consider the maps

$$f_*\mathcal{O}_Y \longrightarrow (X_y \to Y)_*\mathcal{O}_{X_y} \longleftarrow y_*\mathcal{O}_{\mathrm{Spec}(k)}$$

The fibre product is a quasi-coherent sheaf of $\mathcal{O}_Y$-algebras $\mathcal{A}'$ and we can define $X' \to Y$ as the relative spectrum of $\mathcal{A}'$ over Y, see Morphisms, Lemma 01SA. This construction commutes with arbitrary change of base. In particular, it is clear that over the open subspace $|Y| \setminus \{y\}$ the morphism $X \to X'$ is an isomorphism and over $|Y| \setminus \{y\}$ the morphism $X' \to Y$ is integral. It remains to prove the statements in a small neighbourhood of y. Choose an affine scheme $V = \mathrm{Spec}(R)$ and an étale morphism $\varphi : V \to Y$ such that y is in the image of φ. Then V_y is a closed subscheme of V étale over k, whence consists of finitely many points each with residue field separable over k (see Decent Spaces, Remark 03II). After shrinking V we may assume there is a unique closed point $v = \mathrm{Spec}(l) \to V$ mapping to y with l/k finite separable. We may write $V \times_Y X = \mathrm{Spec}(C)$ with $R \to C$ an integral ring map. The stated compatibility with base change gives us that $U \times_X Y' = \mathrm{Spec}(C')$ where

$$C' = C \times_{C \otimes_R l} l$$

Since $R \to l$ is surjective, also $C \to C \otimes_R l$ is surjective and we see that this is a fibre product of the kind studied in More on Algebra, Situation 08KH (with A, A', B, B' corresponding to $C \otimes_R l, C, l, C'$). Observe that C' is an R-subalgebra of C and hence is integral over R; this proves (1). Finally, More on Algebra, Lemma 0B7J shows that $V \times_X Y' = \mathrm{Spec}(C')$ has a unique point y'' lying over v with residue l (this corresponds with the obvious surjective map $C' \to l$). Thus $X_y \times_{\mathrm{Spec}(k)} \mathrm{Spec}(l)$

has a unique point with residue field l. Since l/k is finite separable, this implies X'_y has a unique point with residue field k, i.e., (3) holds.

To prove the final statement, observe that if Y is locally Noetherian, then R is a Noetherian ring and if f is finite, then $R \to C$ is finite. Then C' is a finite type R-algebra by More on Algebra, Lemma 00IT. This proves that $X' \to Y$ is finite. $\square$